

Advanced (Habanero)

Q1. Simplify the expression $\sqrt{16}$

- (A) 2
- (B) 4
- (C) 8
- (D) 16
- (E) 32

Q2. Simplify the expression $\sqrt[3]{27}$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q3. Which of the following is equivalent to $x^{1/2}$?

- (A) $\sqrt{x}$
- (B) $\sqrt[3]{x}$
- (C) x^2
- (D) x^3
- (E) x^4

Q4. Simplify the expression $\sqrt{64}$

- (A) 4
- (B) 6
- (C) 8
- (D) 10
- (E) 12

Q5. Which of the following is equivalent to $\sqrt[4]{x}$?

- (A) $x^{1/2}$
- (B) $x^{1/3}$
- (C) $x^{1/4}$
- (D) $x^{2/3}$
- (E) $x^{3/4}$

Q6. Simplify the expression $\sqrt{121}$

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 14

Q7. Which of the following is equivalent to $x^{3/2}$?

- (A) $\sqrt{x^3}$
- (B) $x\sqrt{x}$
- (C) $x^2\sqrt{x}$
- (D) $x^3\sqrt{x}$
- (E) $\sqrt[3]{x^2}$

Q8. Simplify the expression $\sqrt[3]{64}$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Open Ended Questions (FRQ)

Q9. Simplify the expression $\sqrt{x^2 + y^2}$ and convert it to rational exponent form.

Q10. Prove that $\sqrt{x} \cdot \sqrt{y} = \sqrt{xy}$ and provide an example.

Chef's Tips

- Tip 1: Emphasize the importance of simplifying radicals and converting between radical and rational exponent form.
- Tip 2: Encourage students to use the properties of radicals to simplify complex expressions.
- Tip 3: Remind students to check their work by plugging in values for the variables.